(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

III Semester, November, 2022

Paper Code: EEC 209 **Subject: Electrical Materials**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 A	Draw the energy band diagram for conductors and semiconductors.	2	CO1
B	What is meant by 'Impregnation' of insulating materials?	2	CO2
C	Define the term 'plastics'?	2	CO2
D	State the factors affecting electrical conductivity.	4	CO1
2 A	Comment on the temperature dependence of resistance of conductors as compared to semiconductors.	5	CO1
B	A coil has a resistance of 18 Ω when its mean temperature is 20 °C and of 20 Ω when its mean temperature is 50 °C. Find its mean temperature when its resistance is 21 Ω.	5	CO1
3A	Classify insulating materials on the basis of temperature.	7	CO2
B	Differentiate between "thermoplastic materials" and "thermosetting materials."	3	CO2
4 A	Lead (Pb) behaves as a superconductor at a temperature of T_c= 7.26 K. If the value of critical magnetic field of lead at T= 0 K is H_c= (8 x 10^5) A/m, find the critical magnetic field of lead at 4 K.	5	CO1
B	Write a short note on 'Ceramics.'	5	CO2